Projects, Papers, and Proteins

Do student iGEM projects predict local academic research in synthetic biology?

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iGEM: a global natural experiment in distributed innovation

- 100k+ participants; 400+ universities; 50+ countries
- Teams design organisms around a local problem
- Projects, parts, and wiki pages are publicly archived
- Runs annually – a repeated cross-section from 2009 to 2025

The iGEM corpus gives us a rare window into how student-level scientific interest maps onto local research infrastructure.

Case Study: Edinburgh 2013

The project: WastED

Engineering *E. coli* to produce bioethanol from industrial waste

Core result: enzyme fusion (Pdc–AdhB) raises ethanol yield $\rightarrow$ published in *ACS Synthetic Biology*, 2014

Local infrastructure:

- Advised by Elfick & French – both already professors at Edinburgh
- Edinburgh ran iGEM teams continuously 2009–2025

12 students; where are they now?

- **4 pursued PhDs** (Edinburgh, Cambridge, UCL)
- **1 faculty** (asst. professor, plant biotech, Krakow)
- **3 moved into biotech industry** (Novonesis, Abbott, Meridian)
- **1 startup founder** (Biotangents Ltd, diagnostics)
- 1 commercialisation director (Stevenage Bioscience Catalyst)
- $\sim$3 no public trace

Are student projects topically embedded in local research?

Inspired by:

- Hidalgo et al. (2007): revealed comparative advantage and product space
- Neffke & Henning (2013): skill-relatedness from labor flows

Applied here to text, not products or occupations.

- Unit: city
- Similarity: cosine distance in embedding space
- Question: do student projects and local papers occupy related regions of semantic space?

Core hypothesis

Innovation follows a local path:
student projects → papers → patents

If so: project topics should *lead* paper topics in the same city by a few years.

Data

Corpus	Docs	Cities	Years	Source
Synbio papers	10,402	1,218	1964–2026	OpenAlex
iGEM projects	4,614	626	2009–2025	iGEM Registry
BioBrick parts	89,060	472	2009–2025	iGEM Registry
Analysis sample	–	**387**	–	cities with both

Carbon-capture tagged: 294 papers (2.8%) and 141 projects (3.1%).

SPECTER2 embeds scientific text into a citation-informed space

- 768-dim vectors; contrastive learning on citation pairs
- **Proximity adapter**: fine-tuned for semantic similarity, not classification
- Cached for all 15,000 documents; fully reproducible

City representation:

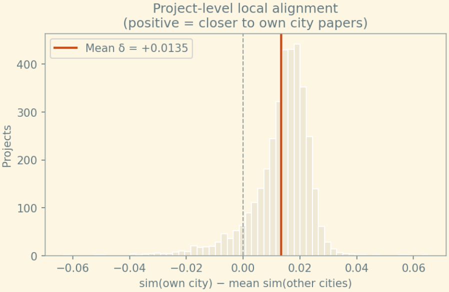
$$\bar{\mathbf{a}}_c = \frac{1}{|P_c|} \sum_{p \in P_c} \mathbf{e}_p$$

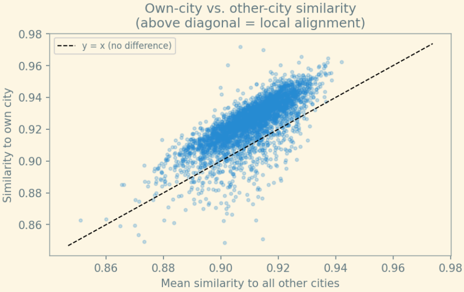
Mean embedding over all docs of type t in city c.

Why SPECTER2?

Standard sentence transformers are not calibrated for scientific text. SPECTER2 is trained on 146M citation pairs; documents that co-cite tend to be nearby.

90% of projects are semantically closer to their own city's papers





n = 3,620 projects; cities with $\geq$ 3 papers; δ = own-city cosine − mean over all other cities

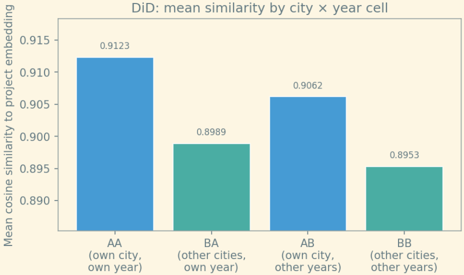
Local Alignment Statistical Tests

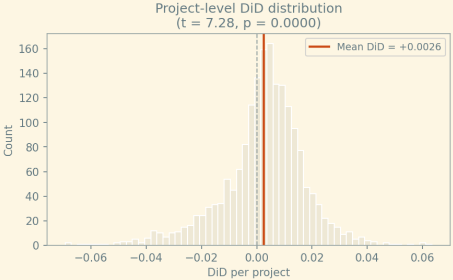
	Statistic	p-value
Mean δ (own − other)	$+0.0135$	
Fraction with $\delta > 0$	90.6%	
One-sample t-test	$t = 78.02$	< 0.001
Wilcoxon signed-rank		< 0.001

Takeaway

Projects are systematically closer to local papers than to papers anywhere else. The effect is small but extremely consistent.

DiD: separating stable specialisation from temporal alignment





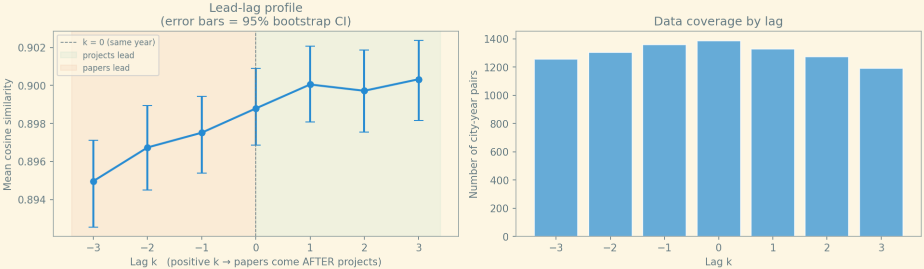
Annual city-year centroids ($\geq$ 2 papers); 1,841 projects in DiD sample

The temporal component is real

	Own city	Other cities	Local advantage
Own year	0.9123	0.8989	$+0.0135$
Other years	0.9062	0.8953	$+0.0109$
Δ (temporal)	$+0.0062$	$+0.0036$	
DiD		$\mathbf{+0.0026}$	

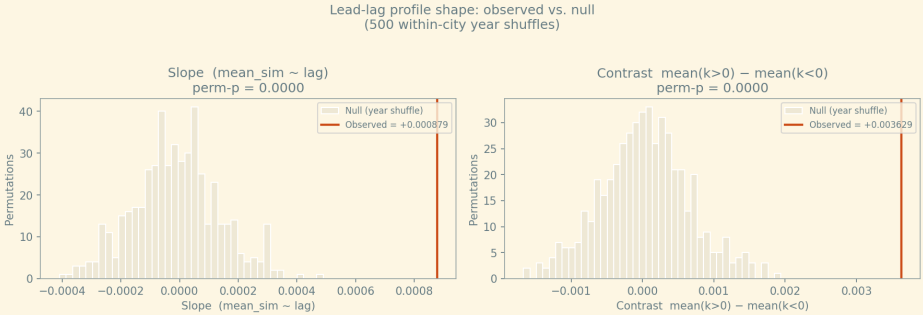
Stable city specialisation accounts for most of the alignment. The DiD component is small but positive.

Projects are semantically ahead of local papers by about 3 years



Lag k: cosine similarity between project centroid in year t and paper centroid in year $t + k$; bootstrapped 95% CI

The lead-lag slope is not a temporal sampling artifact



500 within-city year-shuffles; observed slope $= +0.000879$; null $p_{95} = +0.000246$; perm-$p < 0.05$

UMAP + HDBSCAN finds 83 topic clusters across all 15k documents

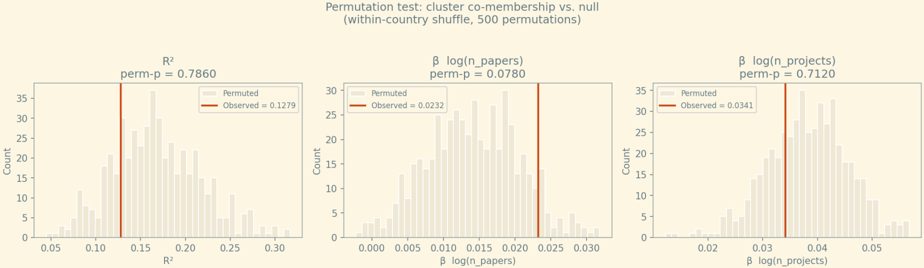
- UMAP: 60 dimensions, cosine metric
- HDBSCAN: 83 clusters, $\sim 20\%$ noise
- Cluster overlap $=$ weighted Jaccard on cluster frequency vectors
- Sample: 189 cities with ≥ 3 clustered docs per type

Mean cluster overlap: 0.081; median: 0.023; max: 0.717

Why this is better

Cluster co-membership is *relative*. It asks: do a city's projects and papers concentrate in the *same topical niches*, not just the same broad field?

Cluster co-membership survives the permutation test



500 within-country shuffles; observed $\beta \ \log n_{\text{projects}} = +0.046^{***}$; perm-$p = 0.03$

Coefficient plot: cluster co-membership is robust

OLS with country FE and robust SE; dependent variable: cluster overlap score

Physical part-type composition independently agrees with semantic clusters

BioBrick parts: standardised DNA sequences registered by teams

- Types: promoter, CDS, RBS, terminator, reporter, device, ...
- City part-type profile: proportion vector over part types
- Shannon entropy measures specialisation

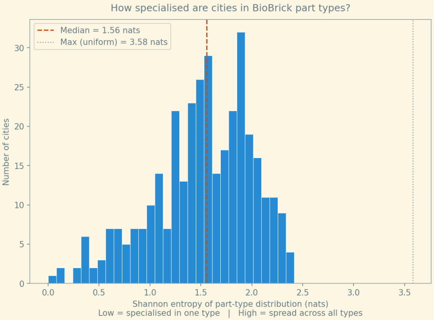
Mantel test: are part-type distances correlated with cluster-space distances?

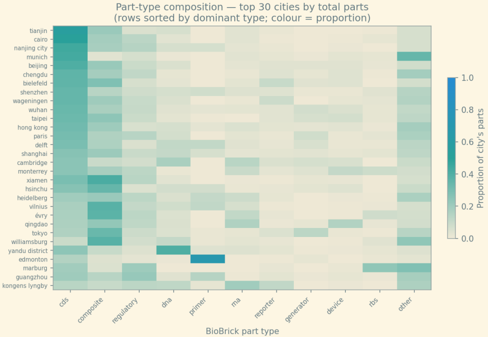
Result (179 cities, 999 permutations)

$$r = 0.2207, \quad p < 0.001$$

Null $p_{95} = 0.073$. Part-type space and semantic cluster space agree.

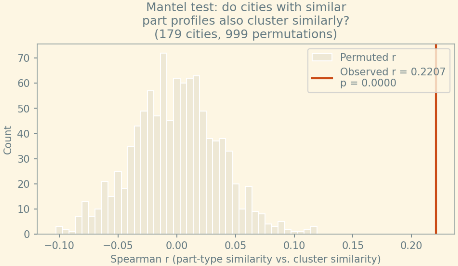
Part-type specialisation varies meaningfully across cities

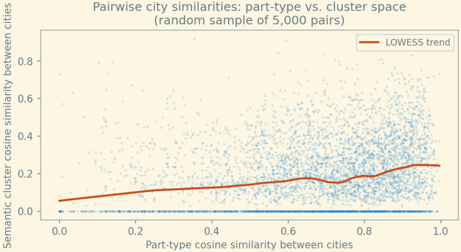




Top 30 cities by total registered parts; sorted by dominant part type

Mantel test: part-type and cluster distances are correlated





Spearman Mantel $r = 0.2207$; perm-$p < 0.001$; 179 cities

1. **Local alignment is real and consistent –** 90.6% of projects are topically closer to their own city's papers ($t = 78$, $p < 0.001$).
2. **Projects lead papers by ≈ 3 years –** lead-lag peak at $k = +3$; profile shape non-random under permutation.
3. **Cluster co-membership is locally specific –** survives within-country permutation where centroid similarity does not ($p = 0.03$).
4. **Part-type space corroborates semantic space –** Mantel $r = 0.22$, $p < 0.001$; two independent signals agree.

Limitations and what comes next

Current limitations:

- No causal identification – shared local environment could drive both
- Temporal relationships difficult to estimate due to unclear research timelines
- iGEM Projects and papers use different language, SPECTER2 not trained on project data
- Patents not yet integrated

Next steps:

- Integrate patent corpus
- Research success; connect to citation count for papers and awards for projects
- Fine-tune model based on links from biobrick registry (paper-part-project)
- Improve interactive semantic space explorer + geographic view

Takeaway

In cities with active iGEM participation, the topics that student teams work on are semantically related to the topics that local researchers publish on – and this co-location in semantic space is not explained by city size alone, holds at the topic-cluster level, and is corroborated by an independent physical signal (BioBrick part-type composition). iGEM projects also tend to share the most similarity with regional research 3 years in the *future*. Causal relationships are not proven; whether this reflects knowledge spillovers, shared local opportunity, or selection of students into research-active cities remains open.